

Restoration-driven positive and negative leakage in sub-Saharan Africa

Received: 20 November 2025

Accepted: 19 August 2026

Published online: 30 September 2026



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Habitat restoration is a cornerstone of global climate and biodiversity policy, with over 350 million ha pledged for restoration by 2030, yet it remains unclear how restoration impacts ‘spill over’ beyond implementation areas. Restoration projects could lead to increased tree loss elsewhere (negative leakage; for example, when restoration replaces timber plantations and/or impacts landowner earnings) or reduced tree loss within project areas and surrounding landscapes (positive leakage; for example, if restoration meets timber or fuelwood resource needs or creates restoration-linked employment). Here we assembled a geospatial dataset of 129,982 restoration polygons (~6.8 million ha) across sub-Saharan Africa to evaluate changes in tree loss across space and time. Natural regeneration and active restoration reduced tree loss by up to 9.1 percentage points (pp) within restoration areas and 8.9 pp in 10 km buffers after 10 years, indicating positive leakage. By contrast, agroforestry increased tree loss by 7.9 pp within projects and 7.5 pp in surrounding landscapes. Leakage magnitude and direction varied across restoration approaches and countries (−7.8 pp to +7.1 pp), underscoring the need for context-specific strategies. Integrating leakage safeguards into restoration planning, carbon standards and incentive schemes is essential for securing landscape-scale climate and biodiversity-related benefits.

Nearly half of the world’s natural habitat extent has been removed, fragmented and degraded, with the pace of loss accelerating in the past few centuries^{1,2}. This has severe consequences for the climate, biodiversity and ecosystem services essential for planetary health and human well-being². Restoring habitats on degraded and deforested land to recover ecosystem functioning and safeguard biodiversity has gained momentum under the UN Decade on Ecosystem Restoration (2021–2030)³. The Bonn Challenge seeks to bring 350 million ha (Mha) of land under restoration by 2030⁴, while pledges made during the UN Climate Change Conference in Glasgow (COP26) committed to investing US\$19.2 billion in halting and reversing forest loss as well

as forest degradation by 2030⁵. There is now a pivotal opportunity to scale up restoration and reverse long-term ecosystem degradation.

Well-managed forest and farmland restoration programmes contribute to carbon sequestration, biodiversity recovery and provision of ecosystem services, including enhanced soil fertility, reduced erosion, improved water quality and rural livelihoods^{6,7}. Current evidence on restoration effectiveness primarily focuses on quantifying outcomes within restored areas⁸. This neglects potential off-site impacts arising from land-use changes that occur beyond restoration boundaries, via displacement (activity leakage) of land use and related environmental impacts that may substantially influence overall restoration outcomes^{9–11}.

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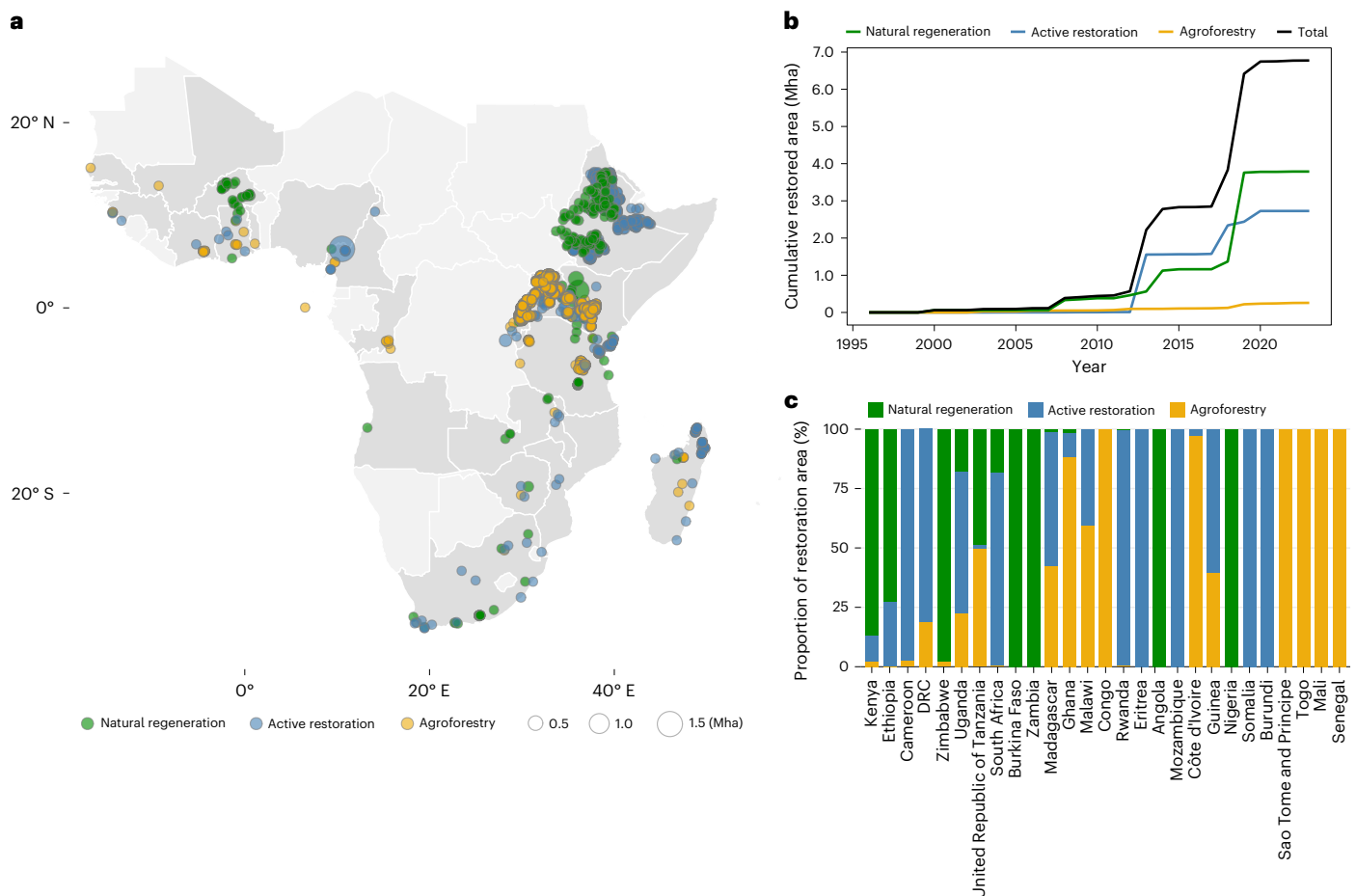


Fig. 1 | Spatial and temporal distribution of different restoration approaches across SSA within our dataset. a, Geographical distribution of clustered restoration sites by approach ($n = 19,229$). Circle size is proportional to the area of each restoration cluster at a given location (in Mha). Countries shown in

darker grey indicate those included in our dataset. **b**, Cumulative restored area from 1996 to 2023 across SSA. **c**, Proportion of restoration area by type for each country in the dataset. Countries are ordered (left to right) by total restoration area within the country.

When restoration replaces productive managed landscapes such as farmland or timber plantations or impacts landowner earnings, it can lead to increased tree loss in nearby tree covers (negative leakage) as people compensate economic losses^{11,12}. In Malawi, rural households engaging in tree-planting projects triggered additional vegetation clearing on their landholdings¹³, while in southwestern China, the Grain-for-Green Program aimed at reforesting cropland between 2000 and 2015 but coincided with a loss of ~7% of native forests in the region¹⁴. Alternatively, where restored forests meet timber and fuelwood resource needs or generate restoration-related employment, it may reduce tree loss (positive leakage)^{7,9,10}. For example, 36 forest restoration and conservation projects in China showed 0.91% to 1.60% forest gain in neighbouring areas¹⁵, while agroforestry interventions across 38 subnational regions in Southeast Asia led to a net reduction in deforestation of 250,319 ha per year within project areas and their surrounding landscapes¹⁶. Such contrasting outcomes highlight the need for a systematic and rigorous understanding of restoration leakage to generate unifying insights about the factors driving positive or negative effects across space and time.

Here we tackle the key issue of how different restoration approaches affect tree cover loss dynamics outside project boundaries, specifically whether restoration results in the unintended displacement of tree cover loss pressures (negative leakage) or contributes to broader landscape-level gains (positive leakage). We focus on sub-Saharan Africa (SSA), a region accounting for nearly half of global restoration pledges^{17,18}, including 129 Mha committed under the AFR100 (the African Forest

Landscape Restoration Initiative) by 2030 (ref. 18) and an expanded pan-African target of over 200 Mha (ref. 17). We addressed three core objectives: (1) to identify the spatial distribution of restoration efforts—spanning natural regeneration, intensively managed active restoration and agroforestry—across SSA, (2) to assess the extent to which these different restoration approaches lead to local-scale increased (negative leakage) or reduced (positive leakage) tree loss over time and (3) to examine how the magnitude and patterns of both types of leakage vary among countries.

Spatial and temporal distribution of restoration in SSA

We compiled a geospatial database of forest restoration activities across SSA, combining 4 sources to identify 129,982 polygons established between 1996 and 2023. These were grouped into 19,229 clusters based on geophysical proximity (<250 m; Methods), spanning 27 countries and covering ~6.8 Mha (Fig. 1a). Natural regeneration (including passive (that is, natural recovery) or assisted natural regeneration; Methods) accounted for the largest share (56.1%, 3.8 Mha), with a marked increase in area of ~3.9 Mha after 2015 (Fig. 1b). Intensively managed, active restoration (Methods) covered 40.2% (2.7 Mha) of the area and showed a steady increase in area over time (0.2 Mha per year on average; Fig. 1b). Agroforestry contributed 3.8% (0.26 Mha) of the total restoration area, with minimal increase throughout the study period (Fig. 1b).

Overall, restoration activity accelerated markedly after 2010, largely driven by the expansion of natural regeneration and active

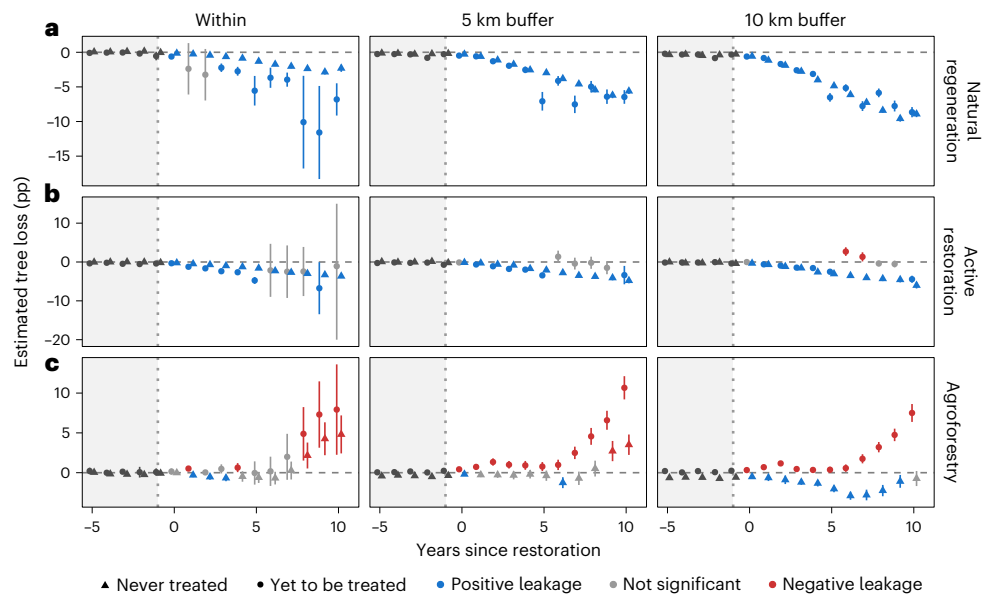


Fig. 2 | Spatio-temporal estimates of cumulative changes in tree loss impacts over time since restoration implementation within the restoration area, 5 km buffers and 10 km buffers. **a–c**, Leakage effects were estimated for three restoration approaches: natural regeneration (analysis unit, sample plot $n = 70,538$) (**a**), active restoration ($n = 98,709$) (**b**) and agroforestry ($n = 46,025$) (**c**). Points represent the ATT estimates of the tree cover loss relative to controls in pp, derived using the DiD estimator proposed in ref. 19, with error bars indicating 95% confidence intervals. Circular points represent estimates based on a yet-to-be-treated approach using plots yet to be restored as controls. Triangular points represent estimates from a never-treated matched counterfactual approach,

where never-treated plots were used as controls selected through statistical matching based on relevant socio-economic and biophysical covariates known to influence deforestation. A value of 0 denotes no detectable difference between treatment and control plots. The x axis denotes years since restoration, including a 5-year pre-treatment period (shown with a darker background and black points) and a 10-year post-treatment period. Colours indicate effect direction: blue = positive leakage effects (reduced tree loss compared with controls, restoration benefits spillover), red = negative leakage effects (increased tree loss compared with controls) and grey = non-significant effects.

restoration (Fig. 1b). The sharp increase in cumulative restoration area after 2010 aligns with the launch of major policy initiatives such as AFR100 and the Bonn Challenge, suggesting that global and regional commitments may have spurred on-the-ground action and/or reporting. The average time since restoration implementation was 6.7 years by 2025, suggesting that most restored areas are still in early successional stages and their long-term ecological performance thus remains unknown; thus, we encourage caution when extrapolating to longer timescales. In terms of country-level distribution, the largest accumulated restoration area was observed in Kenya (2.7 Mha), followed by Ethiopia (1.6 Mha) and Cameroon (1.5 Mha), which together accounted for over 85% of the total restoration area in our dataset (Fig. 1c).

Changes in tree loss within and beyond restoration areas

To assess potential leakage effects of restoration projects, we generated 5 km and 10 km buffers surrounding the restoration clusters and evaluated changes in annual tree loss within restoration areas and their surrounding buffers (0–5 km, 0–10 km) relative to comparable controls. We used two distinct analytical approaches to a before- and after-control intervention framework. The first uses yet-to-be-treated polygons as a space for time control, where the control was selected from areas that were marked as restored years after the restored year of the treatment. The second uses never-treated statistically matched areas as a counterfactual with comparable biophysical and socio-economic conditions (Methods). We then applied a heterogeneity-robust difference-in-differences (DiD) model to detect relative tree loss dynamics over time and across spatial buffers¹⁹. This revealed substantial variation across restoration approaches (Fig. 2), though minor deviations from the parallel trends assumption were observed at small temporal scales.

A total of 18 countries remained for the DiD analysis after filtering for restoration cultures ($n = 10,038$) located in forests or

other woody landscapes, defined as areas with more than 20% tree cover (Supplementary Fig. 1). We also applied alternative thresholds of 33% and 50% tree cover, tree cover and annual loss records from the Global Forest Change dataset²⁰ (see Methods and Supplementary Information for the details of the sensitivity analyses and Supplementary Figs. 2–5). For natural regeneration, treatment areas consistently experienced lower tree loss compared with controls under both analytical approaches (Fig. 2a). Using yet-to-be-treated counterfactuals, restored areas had 6.8 percentage point (hereafter, pp; 95% CI = 4.7–9.0 pp) less tree loss in the 10 years following the start of restoration, while the 5 km and 10 km buffers had tree loss reductions of 6.5 pp (5.4–7.5 pp) and 8.7 pp (7.9–9.4 pp), respectively. Under a never-treated counterfactual, tree cover loss was reduced by 2.3 pp (1.8–2.9 pp), 5.6 pp (5.2–6.0 pp) and 8.9 pp (8.5–9.4 pp) within the restoration area and the 5 km and 10 km buffers, respectively. Thus, reductions in tree loss occurred both within the restored boundaries and the 5 km and 10 km buffers.

Positive leakage was also observed for active restoration, with reductions in tree loss detected in the years immediately following restoration (Fig. 2b). Using yet-to-be-treated counterfactuals, there was a 4.7 pp (3.9–5.6 pp) reduction in tree loss within 5 years of restoration, although no statistically significant changes were observed thereafter, while the 5 km and 10 km buffers had reductions in tree loss of 3.4 pp (2.8–4.1 pp) and 2.5 pp (2.1–2.9 pp) after 5 years, respectively. The matched counterfactuals method showed a 3.6 pp (3.3–4.0 pp) reduction in tree loss within the restored area over a 10-year period, while reductions of 4.7 pp (4.1–5.5 pp) and 6.0 pp (5.2–6.8 pp) were observed in the 5 km and 10 km buffers, respectively. A sharp shift for the yet-to-be-treated effects in the declining tree loss trend after year six of active restoration likely reflects reduced sample availability, as most restored sites were young and few had data for analysis beyond 6 years. The overall trajectory of tree loss change continued to show a declining trend over time.

The positive leakage effects (that is, beneficial reductions in tree cover loss) associated with natural regeneration and active restoration in certain countries potentially arise through several context-dependent reinforcing mechanisms. First, where restoration activities are well designed and implemented, they may effectively engage local communities, deliver environmental education²¹ or improve land tenure²². Such activities can encourage the adoption of conservation practices in nearby areas, including beyond formal project boundaries²³. In some cases, the visible success of restoration efforts may motivate other community members within the same village to participate voluntarily in restoration programmes, amplifying the impact across the landscape²⁴. Second, positive leakage may occur where restored woody land begins to supply ecosystem goods and services—such as fuelwood, fodder or non-timber forest products^{7,25,26}—that are depleted in highly used landscapes, reducing pressure on surrounding natural forests. Third, restoration may reduce incentives to expand agriculture into adjacent forested areas through improving local conditions, such as microclimate or soil quality, thereby enhancing the productivity of existing farmland²⁷, or may bring job opportunities and financial incentives (for example, carbon payments or subsidies), decreasing reliance on forest clearance as a livelihood strategy^{28,29}. These mechanisms depend strongly on local governance, tenure security, stakeholder participation and benefit-sharing arrangements and should therefore be interpreted as possible drivers of positive leakage where it occurs, rather than as inherent outcomes of natural regeneration or active restoration.

Agroforestry tended to show the opposite pattern to natural and active restoration, with consistently higher tree loss in treatment areas (within the restored area and both buffers) compared with controls beginning ~5 years after restoration, indicating negative leakage (Fig. 2c). Using the yet-to-be-treated counterfactual, after 10 years, tree loss increased by 7.9 pp (2.3–13.6 pp), 10.7 pp (9.2–12.1 pp) and 7.5 pp (6.4–8.6 pp) within the restoration area and the 5 km and 10 km buffers, respectively. Using matched counterfactuals, we observed increases of 4.8 pp (2.4–7.2 pp) and 3.5 pp (2.2–4.8 pp) in tree loss within the restoration area and 5 km buffer after 10 years of restoration, respectively. At the 10 km buffer, the never-treated matching approach detected a temporary reduction in tree loss, peaking at 2.9 pp (2.4–3.4 pp) around 6 years post-restoration. Unlike other comparisons, the matched controls already showed significantly lower rates of tree loss up to 5 years before restoration began, suggesting a potential selection bias—that is, restoration may have been preferentially implemented in areas with already declining tree loss trends, which could inflate estimated off-site benefits.

Negative leakage from agroforestry may arise when it is implemented on active, productive cropland, where added tree cover can reduce yields or usable area through shading and root competition³⁰. To compensate for any yield losses, farmers may clear nearby forest or fallow land. Commercially oriented agroforestry systems—such as cocoa, coffee or rubber—can also raise land values or shift labour towards tree cover-clearing activities for short-term gains^{31–33}. Such negative phenomena are more likely to occur in areas where land tenure is unclear, compensation is inadequate or restoration projects lack long-term monitoring³⁴. By contrast, Teo et al.¹⁶ found that agroforestry reduced tree loss in surrounding landscapes in Southeast Asia. Such differences may reflect regional contrasts in land governance, market access or implementation models—agroforestry in Southeast Asia is often embedded in long-standing community forestry systems or supported by strong tenure arrangements, whereas many SSA contexts may relatively lack such enabling conditions^{35,36}. Although we did not examine these local factors directly, previous studies have highlighted their critical role in shaping agroforestry outcomes at the local scale^{37,38}.

While leakage effects generally increase over time, their timings differed among restoration strategies. Natural and assisted regeneration often reduce deforestation within 1–2 years, likely owing to

immediate protection measures (for example, fencing, fire management) and community engagement^{7,39}. By contrast, agroforestry shows delayed negative leakage, with elevated tree loss typically emerging in surrounding areas after 6 years. Early on, agroforestry systems remain productive—trees are small and competition is minimal—but as trees mature, shading and resource competition could reduce crop yields especially in the short term⁴⁰, prompting clearance elsewhere. Unlike natural regeneration and active restoration, which target overharvested land, agroforestry often occurs on actively farmed plots with higher opportunity costs. Thus, both biophysical and socio-economic constraints likely contribute to delayed negative leakage in agroforestry contexts.

Variation in restoration-induced leakage among countries

We included 12 of 18 countries in our dataset for country-level DiD analysis based on the sample size of the available data. Country-specific estimates of cumulative change in tree loss impact varied substantially across restoration approaches and spatial scales (Fig. 3). Positive leakage was consistently observed in Kenya across all restoration approaches and spatial buffers, with the largest effect seen under active restoration, resulting in a 4.0 pp reduction in tree loss at the 5 km buffer after 5 years (95% CI, 2.6–5.0 pp; estimates based on yet-to-be-treated controls; see Supplementary Table 1 for never-treated results and Supplementary Fig. 6 and Supplementary Table 2 for after 10 years results). Tanzania also showed positive leakage for all restoration types, with the strongest effect under active restoration at the 5 km scale (reduced 11.6 pp, 13.6 pp to –36.9 pp). In addition, Rwanda, Cameroon, Madagascar and Nigeria showed positive leakage under active restoration, with the extent of reduction in tree loss generally increasing with distance from the restored area, except in Cameroon.

Negative leakage was observed in several countries. In Ethiopia, after 5 years of restoration, tree loss increased by 5.6 pp (2.4–8.9 pp) within the project area and by 0.6 pp (0.03–1.2 pp) in the 5 km buffer around areas restored through active restoration, while no significant leakage was found for natural regeneration or agroforestry. In Ghana, negative leakage was detected in the 10 km buffer under both natural regeneration (2.1 pp, 0.6–3.5 pp) and active restoration (7.1 pp, 1.8–12.4 pp). South Africa showed a particularly strong negative leakage effect, with a 52.4 pp (51.4–53.3 pp) increase in tree loss within areas restored via active restoration.

Some countries showed variation in leakage outcomes between restoration types. In Uganda, active restoration caused strong positive leakage, with a 10.2 pp (8.9–11.5 pp) decrease in tree loss within restored areas, while natural regeneration approaches revealed minimal changes within the 10 km buffers, whereas agroforestry increased tree loss by 1.6 pp (0.1–3.0 pp) and 0.8 pp (–0.4 pp to 2.0 pp) at the 5 km and 10 km buffers, respectively. In DRC, active restoration led to a 1.5 pp (0.7–2.2 pp) reduction in tree cover loss within restored areas but a 0.9 pp (0.1–1.7 pp) increase at 5 km, while agroforestry showed consistent negative leakage, increasing tree cover loss by up to 2.1 pp (1.2–3.0 pp).

This duality suggests that the effectiveness of restoration depends not only on the implementation approach but also on the geographic and socio-institutional context. For instance, agroforestry systems differ in tree density and crop integration, while active restoration projects vary in planting designs and species composition^{39,41}. Beyond these technical differences, restoration outcomes are strongly shaped by legacies of land-use history, socio-economic pressures and institutional capacity. Historical land use can leave ecological imprints such as soil degradation or habitat fragmentation⁴², while socio-economic conditions—including market incentives, livelihood dependence and community norms—affect local feasibility⁴³. Institutional factors such as tenure security, governance frameworks and enforcement capacity are equally critical, as unclear or poorly enforced land ownership can drive

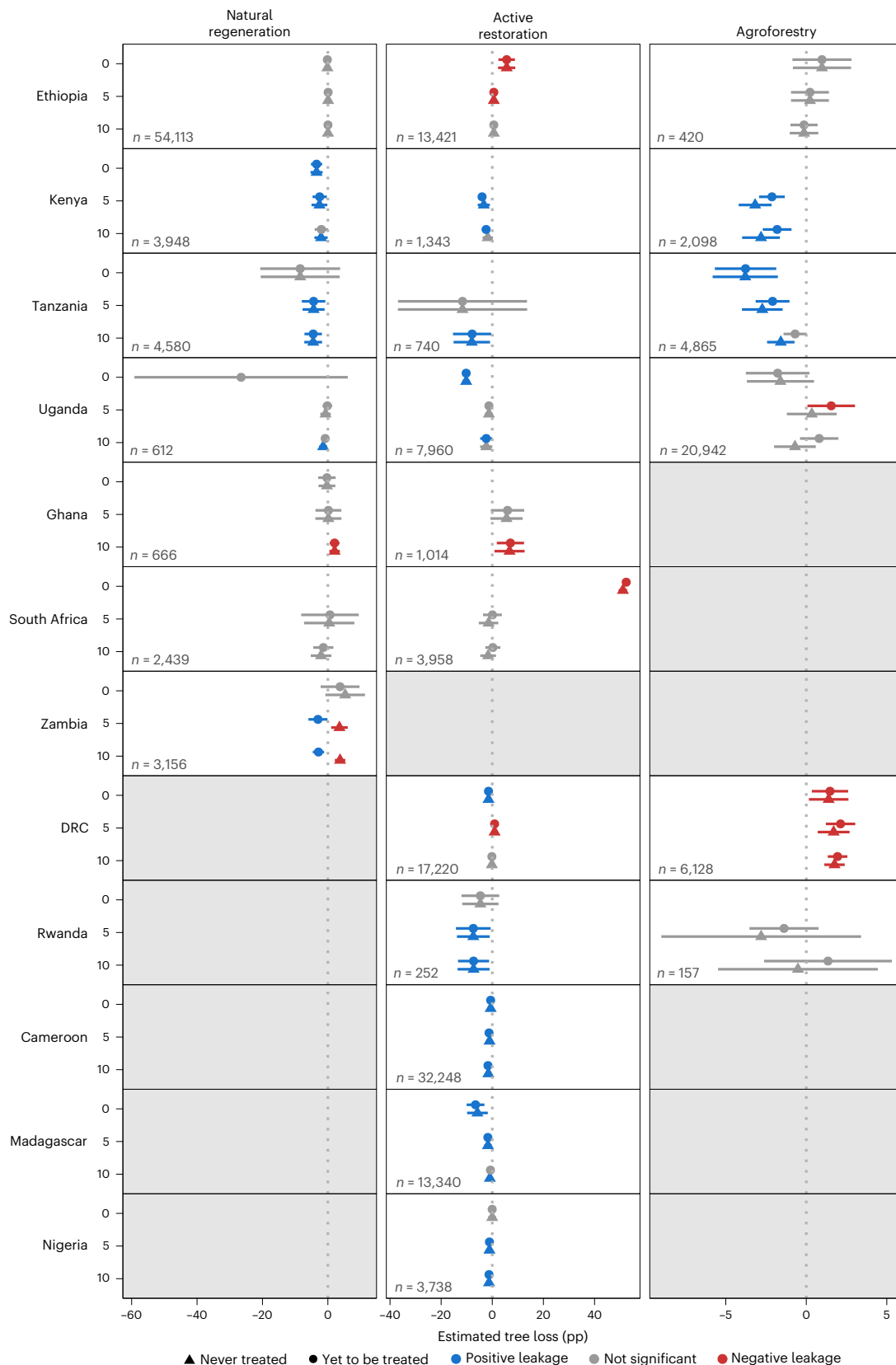


Fig. 3 | Country-specific estimates of cumulative tree loss impact under different restoration approaches (after 5 years of restoration). Estimated tree loss is shown at varying distances from restoration sites (0 = within the restoration area, 5 = 5 km buffer and 10 = 10 km buffer). Points represent ATTs, with error bars indicating 95% confidence intervals. Circles denote estimates based on the yet-to-be-treated approach; triangles denote estimates based on the never-treated matched counterfactual approach. Colours indicate effect

direction: blue = positive leakage effects (reduced tree loss compared with controls, restoration benefits spillover), red = negative leakage effects (increased tree loss compared with controls) and grey = non-significant effects. Grey-shaded regions indicate country-restoration approach combinations for which no estimates were available because no eligible sample plots were present. The sample size of the analysis unit (sample plot) for each restoration type in each country is presented in the bottom-left corner of each box.

opportunistic clearing or encroachment near restoration sites^{22,34}. For example, restoration outcomes in Ethiopia depend strongly on governance and implementation conditions rather than on restoration type alone^{44,45}. Several challenges can undermine restoration effectiveness, including short-term planting campaigns that are poorly matched to restoration timescales, insecure land tenure, limited livelihood support, weak stakeholder participation, poor fit to local contexts and weak coordination across governance levels^{44,45}. These constraints may reduce the long-term effectiveness of both active restoration and assisted natural regeneration, helping to explain why similar restoration approaches can generate very different leakage outcomes across countries and contexts.

Such contextual variation is evident in contrasting agroforestry experiences across Africa. In the Democratic Republic of Congo, collective agroforestry schemes struggled partly owing to widespread poverty and livelihood pressures: farmers under economic stress prioritized work on their own plots rather than contributing labour to communal cocoa-agroforestry fields⁴⁶. By contrast, the Vi Agroforestry programme in Kenya achieved positive outcomes, where individually managed systems improved household income, fuelwood access and milk yields—particularly among female-headed households²⁵. Together, these cases illustrate that land tenure security, incentive structures and participation modes could also fundamentally mediate the direction of restoration leakage effects, determining whether they reinforce or undermine local socio-economic resilience.

Accounting for leakage in restoration programmes

Our findings underscore the urgent need to explicitly account for leakage in restoration-related policy and finance mechanisms, particularly carbon markets. Here we focus on key dimensions for improving restoration outcomes: aligning market and finance mechanisms with net landscape-scale impacts, strengthening leakage accounting and certification standards, and embedding restoration within broader landscape-scale planning and governance frameworks. This is critical for avoiding the underestimation of the positive effects of native forest/woodland restoration approaches, especially in terms of the indirect benefits to surrounding high-conservation-value habitats⁷. Quantifying the wider extent of restoration benefits might help to improve the case for investing in natural regeneration and active restoration by revealing their broader ecological value beyond the restoration site. Our estimates are most informative for medium-term leakage dynamics following restoration establishment. Because most projects in our sample are relatively young and the number of observations declines at longer post-restoration horizons, uncertainty increases at later event times. This is particularly important when considering the longer-term climate benefits of restoration and its relevance to carbon-market applications.

Integrating leakage considerations can also help agroforestry, which covers 60 Mha under AFR100 commitment, 48% of Bonn Challenge pledges⁴, and supports 1.2 billion people worldwide but shows evidence of negative leakage in our study. Incorporating leakage considerations into carbon certification standards and landscape planning is thus crucial not only for improving transparency but also for ensuring rigorous and accountable quantification of leakage in assessing net climate and biodiversity benefits. Our results echo the importance of landscape-scale planning and governance: stronger enforcement of land-use zoning, integration across agricultural and forestry sectors, and embedding restoration within broader spatial planning frameworks are all critical to amplify positive impacts and avoid unintended tree loss elsewhere^{7,8}.

Effective regulatory protection and strong community leadership are critical for sustaining and amplifying the benefits of restoration. Many outcomes—positive or negative—emerge or intensify only after several years, underscoring the need for extended funding cycles,

consistent monitoring and long-term support. Ensuring landowners and communities derive tangible benefits, such as subsidies or sustainable economies (for example, non-timber forest products), helps offset opportunity costs and incentivizes ongoing protection²⁸. Secure land tenure is also essential, as unclear or poorly enforced ownership often drives opportunistic deforestation near restoration sites²². Indeed, a meta-analysis across tropical regions suggested that public land (unmanaged) with unclear land tenure is associated with negative forest outcomes³⁴. Clarifying land rights and involving local communities in restoration design and governance can reduce negative leakage by limiting displacement of land-use pressures. Such inclusive, tenure-sensitive approaches are especially vital in the restoration of landscapes, where ecological and social dynamics are closely intertwined.

This study has several limitations that future research could address. First, while we assembled the most comprehensive georeferenced dataset of restoration polygons in SSA, it remains incomplete. Many projects lack publicly available or independently verifiable spatial data, which means that our dataset represents only a subset of continental restoration efforts and is unevenly distributed across countries. Thus, the pooled estimates should be interpreted cautiously as broad average patterns in the available dataset, rather than as universal effects of restoration across SSA. Second, the Hansen tree cover loss data that we used in this study might have some uncertainties owing to the low canopy density and the prevalence of small-scale disturbances in SSA⁴⁷. Third, our analysis was limited to spatial leakage within a 10 km buffer; thus, we did not assess the potential for broader-scale regional to international leakage given that much of the agricultural or wood fuel-related activity in SSA occurs within smallholder, subsistence-oriented systems with localized markets, while such broader cross-national leakage also needs to be considered for further research. Fourth, we could not model how specific restoration methods (for example, planting density or tree species) or restoration conditions (for example, land-use history) within each restoration approach influenced the magnitude and direction of leakage effects, and in turn, how that was linked to the effect of restoration on a wider range of ecological outcomes, including carbon and biodiversity, soil and water quality, and climate resilience. These impacts warrant further research, especially given that some projects may have been designed primarily to restore soil or hydrological function.

In conclusion, restoration of natural forests or other woody ecosystems, especially through passive natural regeneration or low-intervention, assisted natural regeneration—can generate broad spatial benefits that extend beyond project boundaries. Agroforestry, while widely promoted, appears to present higher risks of negative leakage, although outcomes vary widely across countries and restoration types, reflecting strong spatial and contextual heterogeneity shaped by ecological, institutional and socio-economic factors. Long-term restoration success depends on integrating leakage considerations into certification standards, ensuring secure land tenure and maintaining strong community engagement³⁴. Effective regulatory protection and locally tailored restoration strategies are essential to sustain benefits over time. Funding mechanisms must recognize and reward the ecological value of regenerating native forests, which are often excluded from carbon markets or undervalued in terms of their carbon sequestration potential.

Methods

Geospatial data

The location of restoration areas in SSA was obtained by accessing publicly available datasets and by requesting information from organizations involved in related activities. We included three broad categories of restoration: (1) passive natural or assisted natural regeneration, where natural forests or other woody ecosystems recover with limited or low-intensity assisted intervention, including controlling invasive

species, weed suppression, enhancement of seed dispersal (for example, establishment of ecological corridors), reduced impacts from burning, grazing or wood harvesting and enrichment planting with desirable tree species (herein, natural regeneration)^{39,48}; (2) active restoration, involving intensive interventions or treatment, such as well-designed tree planting with dense native species, to accelerate recovery³⁹; and (3) agroforestry, which integrates trees into croplands to enhance agricultural productivity and resilience while increasing tree cover⁴⁹. For each restoration area, we obtained the start years and restoration approach. We did not include plantations—typically simplified, timber-producing systems—despite their importance as a restoration strategy in many national plans and commitments. This exclusion was due to both the lack of reliable establishment data and the ambiguity in existing datasets⁵⁰ about whether plantations were established for ecological restoration or commercial purposes, limiting our ability to confidently include them as genuine restoration interventions.

We have used data on restoration area locations from the following sources: (1) The TIST Restoration Program⁵¹, which focuses on increasing tree cover by implementing agroforestry approaches. This programme involves farmers planting useful fruit, timber or medicinal plants on their farms. The TIST dataset includes areas restored from 2002 to 2023, mainly located in Kenya, Uganda and Tanzania. (2) The Sustainable Land Management Program, a project hosted by the World Bank⁵², involved over US\$1 billion investment in restoration efforts in Ethiopia. The programme aimed to reduce land degradation and improve land productivity in selected watersheds within targeted regions of the recipient's territory. The restoration activities, conducted from 2008 to 2019, included tree planting and the implementation of physical infrastructure for reducing soil erosion. (3) Restor, a global platform that hosts information on community-led restoration sites self-disclosed by individuals or organizations, incorporates a variety of restoration approaches⁵³. We included projects located in SSA with the categories identified by the Restor database as corresponding to natural or assisted natural regeneration—namely natural regeneration, removal of human disturbances, passive restoration and restoration of natural disturbance regimes. In addition, we incorporated projects classified by Restor as active restoration and agroforestry. All projects included were marked as either completed or ongoing by 2023; proposed areas that had not yet been actively implemented were excluded. (4) The nature-based carbon offset project boundaries. The database was compiled by Karnik et al.⁵⁴ as an open-access global dataset containing the mapped boundaries of nature-based forest carbon projects in 55 countries. The dataset covers multiple intervention categories, including avoided deforestation, improved forest management, afforestation, reforestation and revegetation (ARR). We included only those projects located in SSA that fall under the ARR category defined by Karnik et al. and that align with the restoration types relevant to this study. The carbon offset project dataset primarily consists of restoration initiatives based on natural regeneration, assisted natural regeneration or agroforestry.

When combining restoration boundaries from multiple sources, we manually retained only one boundary per project in cases of spatial overlap. When multiple spatial shapes were available for the same restoration project, we prioritized the most detailed polygon provided directly by the implementing organization or restoration NGO (for example, TIST). These polygons generally offered the most complete and spatially explicit representation of project boundaries, thereby reducing boundary uncertainty and improving consistency in buffer delineation and tree cover loss extraction. We excluded polygons located in wetlands, marine environments or mangroves (mainly from the Restor or nature-based carbon offset project boundaries datasets). This process yielded a final dataset comprising 129,982 polygons, representing around 6.8 Mha of restored land across 27 countries in SSA, with restoration activities conducted between 1996 and 2023.

Data compilation

Given that a large number of the collected restoration polygons were smaller than 5 ha (Supplementary Information and Supplementary Fig. 7) and located in close proximity to one another, we grouped nearby polygons into clusters based on centroid distances of less than 250 m to reduce spatial pseudo-replication and improve ecological interpretability, as proximity may reflect shared ecological characteristics. The clustering was performed using a density-based algorithm from the 'dbscan' package⁵⁵ (1.2.5) in R. Each cluster represents a restoration site, with the earliest identified polygon determining the starting year. The clustering process was conducted separately for each restoration type.

The analysis focused on the impacts of restoration on tree cover. We therefore used Hansen Global Forest Change (GFC) data to filter for restoration clusters that were at least 20% densely forested (defined as >25% tree cover in 2000)²⁰ inside a 5 km buffer. We also used alternative higher thresholds (33% and 50%) to repeat the main analysis for sensitivity analysis. We also excluded sites that were restored before 2000 from our analysis, as the tree cover loss records in the GFC dataset begin in 2000. After dropping 9,191 restoration clusters located outside the forest/other woody ecosystem or restored before 2000, as defined in the preceding, a total sample of 10,038 of the 19,229 restoration clusters remained.

Detecting tree cover loss

To obtain a measure of tree cover loss to compute the leakage driven by restorations, we classified the treatment groups based on proximity to restoration areas as follows: within restoration areas and within areas surrounding the restoration areas in 0–5 km and 0–10 km buffer zones (Supplementary Fig. 8a). We established a grid (that is, a sample plot) with a resolution of 7 × 7 pixels used by the GFC dataset (~200 m) across the target countries and selected grids within a 10 km buffer of restoration sites established each year. Plots overlapping with 10 km buffers around areas restored in other years were excluded from all treatment groups to avoid the impact caused by interventions from other years.

We used the GFC v1.11 (2000–2023) dataset²⁰ in Google Earth Engine (GEE)⁵⁶ to calculate the accumulated annual tree loss before and after the year of restoration. The GFC dataset²⁰ provides information on tree cover and tree loss at 1 arcsec resolution on a global scale. We classified pixels with at least 25% tree canopy cover in 2000 as wooded, following the threshold recommended by ref. 57 for reliably distinguishing tall woody vegetation. Here we used GFC specifically to detect the remaining forest/woody land that surrounds the restoration activities instead of measuring the tree cover loss or gain of the regenerating forest per se. We did not consider tree cover gains as it is substantially more uncertain and the available temporal coverage is not sufficiently long to support a comparable analysis. Then we aggregated the tree cover loss from 1 arcsec to our ~200 m grid cells (7 × 7 pixels per cell) by summing the number of tree loss pixels by their surface intersection with the 7 × 7 pixel cells (that is, the sample plot unit) using GEE (Supplementary Fig. 8). We identified the number of wooded and tree cover loss pixels using the tree loss dataset from 2001 to 2023 for all sampled plots. Sampling plots (7 × 7 pixel cells) were filtered to include only those with at least 1 pixel identified as wooded in 2000, ensuring the plots could potentially experience deforestation activities during the focused time window (2001–2023). Then we calculated the annual cumulative rate of tree loss for sampled plots in each treatment group, and control observations were calculated as:

$$\text{Annual cumulative rate of tree loss (\%)} = \frac{\text{Total number of tree loss pixels since or before restoration}}{\text{Total number of forested pixels in 2000}} \quad (1)$$

Given the large geographic extent and fine spatial resolution of our sampling framework, we randomly selected 25% of the sample

plots generated from each buffer for every restoration cluster from each treatment group within each country, ensuring that we have a fair and representative example across the continent and projects. In total, we obtained 79,389 plots within restoration areas, 88,954 plots for 0–5 km and 199,533 plots for 5–10 km buffers as treatment group pools. The controls pool for the outside treatment zone matching included 779,898 plots across our target countries (details about the controls selection methods in the subsequent Statistical analysis section). The World Mollweide (ESRI, 54009) coordinate reference system was used for this study.

Counterfactual selection

We constructed two groups of counterfactual control plots to support a before-and-after control-intervention analytical framework. The first group consists of treatment plots that are earmarked for restoration but have not yet undergone implementation based on our restoration dataset as a space for time control (that is, the yet-to-be-treated approach). For instance, a plot restored in 2019 could serve as a control for a restoration project implemented in 2009. However, in some countries where recent restoration sites are missing or where restoration occurred in only a single year, this approach results in a limited pool of control plots. To address this limitation, we also used statistical matching to generate counterfactual control plots on lands that are neither designated for future restoration nor affected by protected areas (that is, the never-treated matched counterfactual approach). These plots matched the treatment plots in size and were used to estimate background rates of tree loss. For this second control group, we did not include all available plots. Instead, we selected a subset that closely mirrored the biophysical and socio-economic conditions of the treatment plots to improve comparability (see subsequent Statistical matching section).

Both treatment and control sampling plots were positioned outside of protected areas (PAs). Plots that were located <10 km from PAs were flagged for further statistical matching analyses to account for potential leakage effects of PAs, such as the displacement of tree loss or increased pressure near PA boundaries⁵⁸. We used the UNEP-WCMC boundaries in GEE⁵⁶ to filter plots associated with the PAs. For the second group of control plot sampling, we also excluded plantation areas identified in the Spatial Database of Planted Trees (SDPT Version 2.0) dataset⁵⁰.

Statistical matching

Following previous studies^{15,16,58}, we performed statistical matching⁵⁹ to identify counterfactuals for the never-to-be-restored control group. Selected counterfactuals are comparable to treatments in observable confounders associated with tree loss, including both biophysical and socio-demographic aspects. We conducted matching at the plot-level, using the R MatchIt package (4.7.2)⁶⁰. We measured the similarity between treatment and control points with propensity score matching; the propensity score represents the estimated likelihood of receiving the intervention conditional on baseline characteristics. Each treated observation was paired with the control observation having the most similar score. Covariate balance was assessed using standardized mean differences and was considered acceptable when all standardized mean differences were below 0.25 (ref. 59). We performed 1:1 nearest-neighbour matching with replacement and a caliper size of 0.2 of the standard deviation of covariates to ensure good matches. We included all numeric covariates (slope, elevation, population density, travel time, distance to roads, tree cover and agriculture suitability), with exact matching for PA status, country and ecoregions (more details of matching variables and results can be found in Supplementary Information). We took separate matching for each treatment group. Mahalanobis distance matching was also conducted for each treatment group; however, it was not selected for use in subsequent analyses owing to slightly poorer matching results compared with

those achieved with propensity score matching (Supplementary Fig. 9). All data analyses were performed in R 4.5.2 software⁶¹.

Covariates extraction

We extracted covariates that could influence restoration locations or the risk of tree cover loss. Slope and elevation were obtained from ref. 62, as both factors may affect restoration locations and tree-cover loss⁴⁸. Accessibility was represented by travel time to the nearest settlement with more than 5,000 residents in 2015 from ref. 63 and distance to roads from WorldPop⁶⁴, given the influence of access to markets and transport networks on tree cover loss⁶⁵. Population density in 2015 was obtained from WorldPop⁶⁶ to account for local human pressure on remaining tree cover. Baseline tree cover was derived from the 2010 GFC dataset²⁰ and resampled to 200 m resolution using area-weighted intersections to estimate percentage tree cover in each analytical unit. Agricultural suitability, estimated from climate, soil and topography, was obtained from ref. 67 and included because agriculture is a major driver of tropical deforestation⁶⁸. Finally, country boundaries from the Database of Global Administrative Areas (GADM)⁶⁹ and biomes derived from the WWF Terrestrial Ecoregions of the World dataset⁷⁰ were used for exact matching. These variables accounted for broad national and ecological differences between treated and control observations. We acknowledge that the land-use history of restoration sites could influence subsequent tree-cover-loss dynamics and would ideally be included as a covariate. However, such information was not consistently available across the study region and period. To partly account for these differences, we incorporated pre-restoration tree-cover-loss patterns into the DiD model (see subsequent section), although residual confounding cannot be entirely excluded.

DiD model

The impact of restoration site establishment on surrounding tree cover was estimated using a DiD framework. To leverage the temporal variation in treatment timing, we used a staggered design, where treatment was defined as the initiation year of restoration activities, specifically the earliest year associated with restoration polygons grouped within a given cluster.

We conducted the analysis at both the regional level (SSA) to provide an overall understanding of the off-site impacts of restoration projects and at the country level to enable more targeted insights. We used both yet-to-be-treated restoration sites and never-treated sites as controls. We used the estimator proposed by Callaway and Sant'Anna¹⁹, which improves upon the conventional two-way fixed effects (TWFE) estimator. The TWFE approach can produce biased estimates in settings with staggered treatment timing and heterogeneous treatment effects, owing to the way it implicitly weights comparisons across groups and time periods. By contrast, the Callaway and Sant'Anna estimator calculates group-time-specific average treatment effects on the treated (ATTs; shown as the following equation), allowing for more flexible and unbiased estimation under treatment effect heterogeneity and variation in treatment timing¹⁹.

$$ATT_{(g,t)} = \frac{1}{N_g} \sum_{i: G_i=g} [Y_{it} - Y_{i,g-1}] - \frac{1}{\tilde{N}_g} \sum_{i: G_i \in \tilde{G}} [Y_{it} - Y_{i,g-1}]. \quad (2)$$

Here N_g denotes the number of restoration cluster observations first treated in year g , whereas g refers to a specific treatment cohort. $\tilde{G}_g$ denotes the set of comparison observations not yet treated by year g , and $\tilde{N}_g$ is the number of such observations. Y_{it} is the annual cumulative rate of tree cover loss for observation i in year t , measured within a sampled buffer belt surrounding a restoration cluster, and $Y_{i,g-1}$ is the same outcome measured in the year immediately before treatment. G_i denotes the first treatment year of restoration cluster i . The estimated ATT for restoration initiated in year g and observed in year t is calculated as the difference in the outcome variable Y between year t and year $g-1$,

averaged across all restoration areas established in year g , minus the corresponding average difference for areas that not yet treated by year g , denoted by $\bar{c}$. We then aggregated these group-time-specific ATTs relative to the timing of treatment to produce dynamic ATT estimates.

Similar to the TWFE estimator, identification of causal effects using this method hinges on the assumption of parallel trends between treated and control units in the absence of treatment. That is, if tree cover loss would have followed similar trajectories in both groups without restoration, the ATT can be interpreted as the causal effect of restoration on tree loss⁶⁹. To test parallel pre-trends, we computed pseudo-ATTs, also referred to as placebos, for pre-treatment periods. These placebo estimates use the same specification as equation (2) but are applied to time periods before the intervention. If these estimates are statistically insignificant and close to 0, it supports the claim that pre-treatment trends between treatment and control units are indeed parallel¹⁹.

Sensitivity analyses

To assess the robustness of our findings, we conducted three sets of sensitivity analyses. First, we repeated all analyses using more stringent tree cover thresholds, retaining only restoration clusters with at least 33% and 50% tree cover within a 5 km buffer. Second, we re-estimated the DiD models with additional biophysical and socio-economic covariates to adjust for potential residual confounding. Third, to address potential year-to-year uncertainty in the Hansen tree cover loss data, particularly that arising from cloud cover and related classification noise, we reran the analysis using a 3-year moving average of annual tree cover loss. Specifically, for each focal year t , we replaced the original annual loss value with the mean of loss observed in years $t - 1$, t and $t + 1$. We then recalculated cumulative tree cover loss and repeated the DiD analysis using these smoothed values. This approach reduces the influence of short-term fluctuations and spurious interannual variation in the remote-sensing product while preserving broader temporal trends in tree cover loss dynamics. Our results are robust, with both sensitivity analyses yielding patterns consistent with the main findings (Supplementary Figs. 2–5).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Publicly available data supporting the findings of this study are provided in the Methods section and in Supplementary Information, which include full source details. Source data are provided with this paper.

Code availability

The code for the analysis is available via GitHub at https://github.com/xinran-miao/Miao_et_al_restoration_leakage_SSA.git.

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Acknowledgements

We thank A. Balmford for his valuable suggestions and J. P. G. Jones for her constructive comments on the statistical methods. We are also grateful to W. Ye and Y. Chen for technical support with computing

resources and to C. Wheeler and all institutions and organizations that provided valuable resources and boundary data for this dataset.

Author contributions

X.M., D.P.E. and C.M.R. conceived the study. X.M., O.M. and C.G.B. developed the methodology. X.M. performed the analysis and drafted the original paper. X.M. collected the data with the help of C.M.R., D.P.E., L.K.W., S.M.T. and T.W.C. All authors contributed to reviewing and editing the paper.

Funding

This work was supported by the Natural Environment Research Council (grant number NE/Z50404X/1), awarded to D.P.E. and C.M.R. O.M. was supported by the Leverhulme Trust (ECF-2024-005), and L.K.W. by a Google Carbon Removal Research Award.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41893-026-01936-2>.

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Peer review information *Nature Sustainability* thanks Dula Wakassa Duguma and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

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Data collection	We used R 4.5.2 , Google Earth Engine to process the data. Full details and references are available in the Methods and Supplementary Information.
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Describe the covariate-relevant population characteristics of the human research participants (e.g. age, genotypic information, past and current diagnosis and treatment categories). If you filled out the behavioural & social sciences study design questions and have nothing to add here, write "See above."

Recruitment

Describe how participants were recruited. Outline any potential self-selection bias or other biases that may be present and how these are likely to impact results.

Ethics oversight

Identify the organization(s) that approved the study protocol.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☐ Behavioural & social sciences ☒ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description

We assembled a geospatial dataset of 129,982 restoration polygons (6.8 million hectares) across sub-Saharan Africa, capturing natural regeneration, active restoration and agroforestry projects, to evaluate changes in tree loss across space and time by using Difference-in-Differences (DiD) framework estimated average treatment effects on the treated.

Research sample

A candidate 129,982 restoration polygons. In total, we obtained 79,389 sampled plots within restoration areas, 88,954 plots for 0-5 km, and 199,533 plots for 5-10 km buffers as treatment group pools. The controls pool for the outside treatment zone matching included 779,898 plots across our target countries.

Sampling strategy

To assess potential leakage effects of restoration projects, we generated 5 km and 10 km buffers surrounding the restoration clusters and evaluated changes in annual tree loss within restoration areas and their surrounding buffers (0-5km, 0-10km) relative to comparable controls. We employed two distinct analytical approaches to a before- and after-control intervention framework. The first uses yet-to-be-treated polygons as a space for time control where the control was selected from areas that were marked as restored years after the restored year of the treatment. The second uses never-treated statistically matched areas as a counterfactual with comparable biophysical and socioeconomic conditions (see Methods).

Data collection

Data were downloaded from open-access sources or obtained from authors.

Timing and spatial scale

We collected restoration area established from 1996-2023 and conducted the DiD analysis for restoration site established after 2000 by using the forest loss data from Hansen (2001-2023). We detail in the Methods and Supplementary Information the source and version of data used in this study. All data used are at the sub-Saharan Africa.

Data exclusions

The analysis focused on the impacts of restoration on tree cover. We excluded restoration sites that located outside forests or other woody landscapes. We also excluded sites that were restored before the year 2000 from our analysis, as the tree cover loss records in the GFC dataset began in 2000.

Reproducibility	This study does not involve experiments and is reproducible from code.
Randomization	Given the large geographic extent and fine spatial resolution of our sampling framework, we randomly selected 25% of the sample plots generated from each treatment group within each country to reduce the computational burden. In total, we obtained 79,389 plots within restoration areas, 88,954 plots for 0-5 km, and 199,533 plots for 5-10 km buffers as treatment group pools. The controls pool for the outside treatment zone matching included 779,898 plots across our target countries (details can be found in the Methods).
Blinding	Blinding was not relevant to this study as this was not an experimental study.
Did the study involve field work?	☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.
Novel plant genotypes	Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.
Authentication	Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.